# Ankush Pratap Singh

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Google Scholar

## Education

**New York Institute of Technology** **New York, US**
*PhD in Computer Science, CGPA 4.0/4.0* *Sept 2024 - Present*

**New York University** **New York, US**
*Master of Science in Computer Engineering, CGPA 3.87/4.0* *Sept 2021 - May 2023*

**Netaji Subhas Institute of Technology** **New Delhi, India**
*Bachelor of Engineering in Instrumentation and Control, CGPA 3.60/4.0* *Aug 2013 – June 2017*

## Research Experience

**PhD Researcher,** New York Institute of Technology, New York, USA Sept 2024 – Present

*Project: Multimodal Interaction Learning for Emotion Recognition and Reasoning*

- Advancing multimodal learning techniques for emotion recognition and reasoning, with a focus on modeling cross-modal interactions, information decomposition, and curriculum-based training strategies to improve generalization across heterogeneous modalities.
- Designing and validating experimental frameworks to characterize sample and interaction difficulty, and developing adaptive training strategies that evolve with model learning state.
- Collaborating with research groups at New York University, contributing to a multi-institutional effort for human-centered multi sensory AI applications.

**Research Assistant,** Ai4CE Lab, New York University, New York, USA Sept 2022 – Oct 2023

*Project: Human-Inspired Summarization: Cluster Scene Videos into Diverse Frames*
*(Accepted for Oral Presentation ICCVW 2025)*

- Developed a self-supervised contrastive learning framework to summarize large-scale scene datasets by selecting representative images that maximize spatial diversity and coverage of scenes.
- Performed comparative analyzes of established video summarization methods using the Habitat and KITTI datasets, showcasing the benefits of spatially diverse summarization.
- Engaged in the entire research cycle, including dataset design, preprocessing, model development, performance evaluation, and drafting and revising the research paper.

**Researcher Assistant,** Video Lab, New York University, New York, USA Sept 2022 – May 2023

*Project: Segmenting Metastatic Brain Tumors Using Deep Learning (M.S. Thesis)*

- Investigated deep neural network architectures for medical image segmentation, focusing on metastatic brain tumors using the NYUMets dataset.
- Proposed and evaluated temporal-aware segmentation methods by enhancing UNet with LSTM and Transformer based modules to capture sequential image dependencies, demonstrating the advantages of spatio-temporal modeling for medical imaging tasks, including clinical tumor monitoring and treatment planning.
- Conducted extensive evaluations of segmentation performance using established medical imaging metrics, including Dice Score, Hausdorff Distance, per-class IoU, tumor volume, tumor count, and F-beta scores.

## Teaching Experience

**Graduate Course Assistant,** New York Institute of Technology, New York, USA Sept 2024 – Present

*Courses: Database Management, Big Data Analytics*

- Guided students in applying database concepts and big data frameworks (SQL, Hadoop, Spark) to real-world case studies.
- Designed lab exercises and assignments emphasizing data modeling, distributed processing, and scalability.

**Graduate Course Assistant,** New York University, New York, USA Feb 2022 – May 2023

*Courses: Machine Learning, Real-Time Embedded Systems*

- Assisted faculty in course content preparation, grading, and exam evaluation.
- Supervised student projects through discussion sessions and office hours, guiding implementation and troubleshooting to help translate theoretical concepts into practical solutions.

## Industry Experience

**Software Engineer,** GO-MMT (Goibibo Group), Gurgaon, India Jan 2019 – Apr 2021

*Technologies: Python, Data Science, Full-Stack Web Developement, SEO, Git*

- Engineered end-to-end hotel vertical platform, building back-end services (Kafka pipelines, MongoDB storage) and optimizing front-end image workflows, improving hotel image visibility from 78% to 97%.
- Applied data-driven insights into booking behavior and search patterns to refine personalization, enhance user engagement, and boost SEO rankings across hotels, trains, buses, and destinations.
- Designed and deployed scalable REST APIs and dashboards, collaborating with product, mobile, and marketing teams to launch domestic and international travel platforms from scratch.

**Software Engineer,** Bhavna Software India Pvt. Ltd., Noida, India Jun 2017 – May 2018

*Technologies: Angular 2, TypeScript, C#, .NET, SQL, Git*

- Developed and maintained full-stack features for lease management modules, including CRUD operations, imports/exports, and record maintenance.
- Optimized database queries and back-end logic, improving system efficiency and data consistency across records.
- Collaborated with cross-functional teams to integrate new features, streamline workflows, and deliver user-friendly front-end interfaces.

## Publications & Thesis

[1] **A. P. Singh**, H. Cao, and Y. Liu, "SPICE: Synergy and Partial Information Based Curriculum Evolution," in *Proc. 28th International Conference On Multimodal Interaction (ICMI), Association for Computing Machinery*, 2026.[Online]

[2] **A. P. Singh**, H. Cao, and Y. Liu, "CHUCKLE - When Humans Teach AI to Learn Emotions the Easy Way," in *Proc. Interspeech*, 2026.[Online]

[3] C. Chen, **A. P. Singh**, M. Zhu, Y. Yan, F. Juefei-Xu, and C. Feng, "Human-Inspired Summarization: Cluster Scene Videos into Diverse Frames," in *Proc. 2nd Workshop on Human-Inspired Computer Vision (HCV), ICCVW*, 2025.[Online]

[4] **A. P. Singh**, "Segmenting Metastatic Brain Tumor Using Deep Learning," *Master Of Science Thesis, Department of Electrical and Computer Engineering, New York University*, 2023.[Online]

## Technologies

**Programming Languages:** Python, C, C++, SQL

**Machine Learning & Deep Learning:** PyTorch, Scikit-learn, Hugging Face, OpenCV, LangChain, Librosa

**Data & Big Data:** NumPy, Pandas, SciPy, PySpark, Hadoop, Matplotlib, Seaborn

**Tools & Frameworks:** Git, Docker, Django, Kafka, REST APIs, Redis

**Databases & Cloud:** MySQL, PostgreSQL, MongoDB, Hive, AWS

**Operating Systems:** Linux, macOS, Windows